Jeff Johnson

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Employment

SpaceXAI, Bellevue, WA [January 2026-Present]
Supercomputing – Member of the Technical Staff – IC/Manager

- Built greenfield VM-based neocloud infrastructure for GB300 tenants with overlay networking offloaded to the Bluefield-3 NIC, live migration, managed Kubernetes offering supporting 15k+ nodes per cluster without replacing etcd, and spot-style compute offering used for the companies batch data processing.
- Resolved long-standing reliability issues, circular dependencies, visibility gaps across the team's ownership stack spanning DNS (e.g serving 500k+ QPS from 5 CoreDNS replicas) to training and inference services (e.g TiB/s cross-datacenter file transfers) by profiling, simplifying, and overhauling brownfield services.
- Introduced consistent operational surfaces, external probing, sources of truth spanning 60+ large scale clusters to drive service improvement that shifted outages from user reported, to paging, to non-reoccuring.

Google Cloud, Seattle, WA [2016-2025]
BigQuery Platform – Senior Staff Software Engineer [2025]

- Set multi-year charter for isolation, security, and reliability for BigQuery. Landed cross-component infrastructure for evolving BigQuery from a serverless multi-tenant exabyte scale data warehouse towards virtualized multi-single tenancy across compute, storage, monitoring, and networking.
- Re-organized multi-cloud infrastructure team for horizontal innovation on GCP/Borg, hired and transitioned management. Defined the new technical strategy for cross-cloud querying on top of cross-cloud interconnect.

BigQuery Omni – Senior Engineering Manager [2021-2025]

- BigQuery Omni was an ambitious project to run Google Cloud's marquee serverless data analytics service on AWS and Azure. Together, we raised BigQuery to be a definitive leader in Lake Analytics from Gartner/Forrester, won key deals for Google Cloud, and built an excellent team.
- Initially supported the infrastructure team as a manager then expanded scope to the product engineering team (30-40 SWE). Created and ran a customer engagement program across product/sales/SWE to drive revenue and improve the service.
- Launched first in the industry capabilities through cross-cloud materialized views, joins, delivered on strategic partnership with Salesforce. Expanded to all major AWS regions, reduced infrastructure costs by $XXM, evolved architecture to reduce ongoing maintenance cost to <2 SWEs.

BigQuery Omni Multi-Cloud Infrastructure – Staff Software Engineer [2019-2021]

- Joined at the beginning of the Omni project, focusing on cloud infrastructure, operations, and architecture for the distributed query engine, distributed RAM/NVMe file system, and soft RDMA on Kubernetes.
- Led cross-organizational efforts and provided key technical contributions to port Container-Optimized OS to AWS/Azure, integrate multi-cloud Terraform into Google rollouts, and expand Anthos Kubernetes for first-party workloads. This was recognized externally as a strategic GCP initiative. (coverage)
- Designed a ground-up Google3-native multi-cloud Kubernetes distribution Integrated with key production services (e.g: LOAS, Monarch), enabling SRE-supported Google3 services on AWS/Azure/GCP for the first time, and facilitated the migration of live production traffic from Anthos to meet aggressive GA timelines.
- Achieved security accolades unbreakable by orange teams, external pentesting, and compliance certifications (SOCS/HIPAA..) by raising the bar in security, solving source transparency, identity, and auditing in cloud.

Google Kubernetes Engine (GKE), Anthos – Senior Software Engineer [2018-2019]

- Founding member of the team that built GKE On-Prem, Google's first external Kubernetes distribution. Drove design and implementation for upgrades, addons, led the GA release for the cluster lifecycle team.
- Designed and developed framework for integrating first party partner teams into Anthos. Led open sourcing and creation of the Kubernetes Working Group for SIG-Cluster Lifecycle. (github, kubecon talk)

Cloud Graphite – Software Engineer [2016-2018]

- Built Kubernetes into Cloud Foundry with BOSH, lead networking strategy and app integration. (github)

Microsoft, Redmond, WA – Software Engineer, Redmond, WA [2013-2016]

- Shipped new charting engine in Office 2016 with a focus on cross platform UI. Built support for collaboration into OfficeArt content model to enable real time editing in Office Desktop and Mobile.

Research

BigQuery's Evolution towards a Multi-Cloud Lakehouse [ACM SIGMOD 2024]

Education

Illinois Institute of Technology, Chicago IL [2009-2013]

- B.S. Electrical Engineering, Computer Engineering. Magna Cum Laude.

Expertise

Programming Languages, Tools

- Golang, Rust (AI generated), Ruby
- Linux, Kubernetes, VMs, Containers, Bazel, Terraform, AWS/Azure/GCP/OCI

Focus Areas

- Happy, healthy, productive, operationally excellent teams
- Large scale distributed systems, networking, operating systems, and their subtle interactions